Business Services Industry Analysis

Financial benchmarking and business disclosure analysis with a Microsoft strategy assessment

BAIT 508 Group Project | Section number: BA2

Team members and responsibilities

Full name	Role
Fan Yang	Data preparation and integration; Microsoft analysis and recommendation
Levia Zeng	Industry selection and financial benchmarking
Tina Tory	Text cleaning, keyword analysis, wordclouds and word2vec

All members contribute to interpretation, review and final documentation. Bracketed contact and section fields must be completed before submission. Confirm each member's official full name.

Executive summary

We analyze Business Services firms in SIC major group 73 using 1994–2025 financial records and 2020 and 2025 Item 1 business disclosures. The financial sample contains 30,610 firm-years across 3,273 firms; matched disclosures cover 477 firms in 2020 and 510 in 2025. Cleaning, keyword counts, TF-IDF and Word2Vec support the language comparison and a financial strategy assessment of Microsoft.

Microsoft has the highest 2020–2025 revenue growth among the three selected software benchmarks, at 14.52% annually. Its recent asset productivity nevertheless weakens: from 2022 to 2025, assets grow 19.27% annually while revenue grows 12.42%, and return on assets (ROA) falls from 19.94% to 16.45%. We recommend milestone-based funding reviews for major asset expansions, using utilization and revenue evidence before subsequent funding is released.

Disclosure language shows continuity with changes in emphasis: seven of the ten most frequent terms recur across years, while data leads both 2025 keyword rankings. Its increased token share persists, more modestly, among continuing firms; the apparent rise in platform language reverses in that sample. These patterns suggest evolving disclosure emphasis, rather than demonstrating industry transformation. Filing identity issues qualify the text findings.

Coverage: Steps A–G are explained in the following sections. All financial levels use source units and fiscal-year labels; ratios are explicitly labeled. The report includes selected code as editable text and exported figures.

Step A Industry scope and data integrity

We choose one industry, Business Services, identified by SIC major group 73. This broad classification includes software and other service businesses; it should not be interpreted as a pure software market. We standardize identifiers and preserve the raw inputs before filtering financial rows by the first two digits of SIC.

```
public_clean["major_group"] = public_clean["sic"].str[:2]
sector_panel = public_clean[
    public_clean["major_group"] == "73"
].copy()
```

Coverage measure	Result
Unique firm-year observations	30,610
Unique firms using GVKEY	3,273
Firms with all 32 years from 1994 to 2025	64

A firm-year is a unique GVKEY and fiscal-year pair. After confirming that years lie within 1994–2025 and that no pair is duplicated, 32 distinct years establishes a complete history. Only 1.96% of firms meet this condition, making composition changes an important consideration for annual averages.

Preparation and filing joins

We trim whitespace, convert blank strings to missing values, pad GVKEY and CIK identifiers, and convert financial measures to numeric values. Financial records must have unique firm-year keys. Missing financial values are retained rather than imputed. ROA is calculated only when assets are positive and income is available.

For Steps D and F, filings are joined to the same labeled fiscal year using CIK, the SEC registrant identifier. The 2020 audit finds 112 GVKEY disagreements, supporting CIK as the join key while retaining financial GVKEY as the company identifier. A one-to-one check stops ambiguous joins. Exact duplicate rows are removed; distinct nonblank disclosures under one CIK are reviewed through a longest-text selection rule with deterministic tie-breaking. Length is a completeness proxy, not proof of the latest or correct filing.

```
matched = filings.merge(
    financial_year, on="cik", how="inner",
    validate="one_to_one"
)
```

Filing audit	2020	2025
Raw rows	5,481	4,536
Exact duplicate rows removed	329	0
Unique registrants retained	4,793	4,509
Registrants without a financial match	674	354
Matched rows with GVKEY disagreement	112	0
Matched Business Services disclosures	477	510

Data qualification. Structural join validation does not verify filing content. One 2020 output labeled AUTOMATIC DATA PROCESSING contains stevia and stevioside as its leading terms, an unexpected cluster requiring source review. Together with the 112 identifier disagreements, this limits confidence in firm-specific text attribution and cross-year sector conclusions. We do not use that row as a business insight.

Step B Financial rankings

We rank nonmissing stock prices separately in 2020 and 2025, using GVKEY to break ties. A high share price does not imply high enterprise value or superior investment performance; share counts, currencies and stock splits can affect comparisons.

Rank	2020 company	Price	2025 company	Price
1	ALPHABET INC	1,752.64	CONSTELLATION SOFTWARE INC	2,399.74
2	CONSTELLATION SOFTWARE INC	1,303.04	FAIR ISAAC CORP	1,690.62
3	SHOPIFY INC	1,131.95	UNITED RENTALS INC	809.32
4	COSTAR GROUP INC	924.28	APPLOVIN CORP	673.82
5	TRADE DESK INC	801.00	INTUIT INC	662.42
6	SERVICENOW INC	550.43	META PLATFORMS INC	660.09
7	FAIR ISAAC CORP	511.04	SPOTIFY TECHNOLOGY SA	580.71
8	ADOBE INC	500.12	MASTERCARD INC	570.88
9	PAYCOM SOFTWARE INC	452.25	CACI INTL INC -CL A	532.81
10	TYLER TECHNOLOGIES INC	436.52	S&P GLOBAL INC	522.59

Alphabet leads the 2020 ranking at 1,752.64 source price units; Constellation Software leads 2025 at 2,399.74. The rankings describe reported price levels, not stock returns.

Highest historical annual sales

We interpret historical sales leadership as each firm’s highest observed annual sales, retain its peak year and rank unique firms. This avoids counting one large company repeatedly and differs from summing revenue over unequal firm histories.

Rank	Company	Peak year	Annual sales
1	ALPHABET INC	2025	402,836.00
2	MICROSOFT CORP	2025	281,724.00
3	META PLATFORMS INC	2025	200,966.00
4	INTL BUSINESS MACHINES CORP	2011	106,916.00
5	ORACLE CORP	2025	67,357.00
6	FUJITSU LTD	2010	54,717.32
7	NEC CORP	1999	48,460.65
8	SAP SE	2025	43,207.82
9	SALESFORCE INC	2025	41,525.00
10	LUCENT TECHNOLOGIES INC	1999	38,303.00

Alphabet, Microsoft and Meta occupy the first three positions. These financial levels remain in source monetary units. Without confirmation of currency and scaling conventions, their differences should not be presented as currency-standardized global market shares.

Step B Geography and financial trends

Location code	Unique firms	Location code	Unique firms
USA	2684	HKG	39
CHN	152	SGP	33
CAN	80	JPN	22
ISR	76	IRL	15
GBR	56	NLD	11

Each firm is assigned its latest nonmissing location within the selected sector panel. The USA accounts for 2,684 firms, approximately 82.0% of the 3,273 firms in the financial sample. Location is a recorded company attribute, not a measure of where revenue is earned.

Average stock prices and the 2008 decline

The arithmetic mean stock price rises from 12.61 in 1994 to 47.77 in 2025. The mean falls from 15.59 in 2007 to 9.47 in 2008, approximately 39.3%. The contributing sample changes from 750 to 690 observations, so this is not a fixed-portfolio return.

Matching 2007 and 2008 prices by GVKEY, requiring a positive starting price and a nonnegative ending price, identifies the row labeled ACCESS NEWSWIRE INC as the largest observed decline: 1.25 to 0.0071, or −99.43%. This meets the brief's price-drop criterion but does not establish that the financial crisis caused the decline or that the supplied company name was used historically.

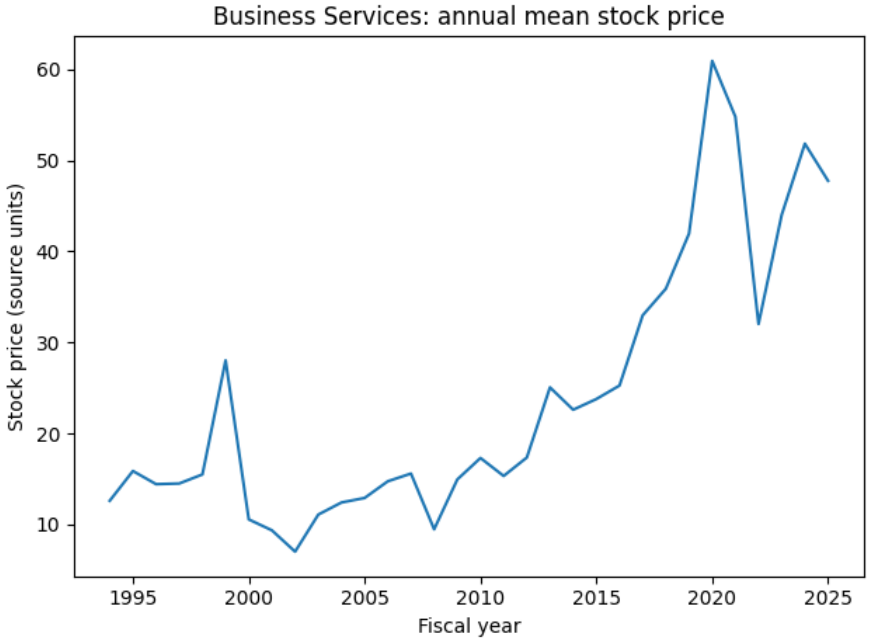


Figure 1 Annual mean stock prices

USA return on assets

We calculate net income divided by assets for each valid USA firm-year, then take the unweighted annual mean. The mean is −18.65% in 1994 and −53.43% in 2025; it reaches −721.76% in 2016. These extreme means can result from large losses relative to very small positive assets and do not describe the typical firm's profitability.

The requested arithmetic mean is retained, with observation counts. A median or trimmed-mean sensitivity analysis would be needed before turning this series into a commercial sector-profitability conclusion. Our Microsoft case uses company-level ratios rather than this unstable industry average.

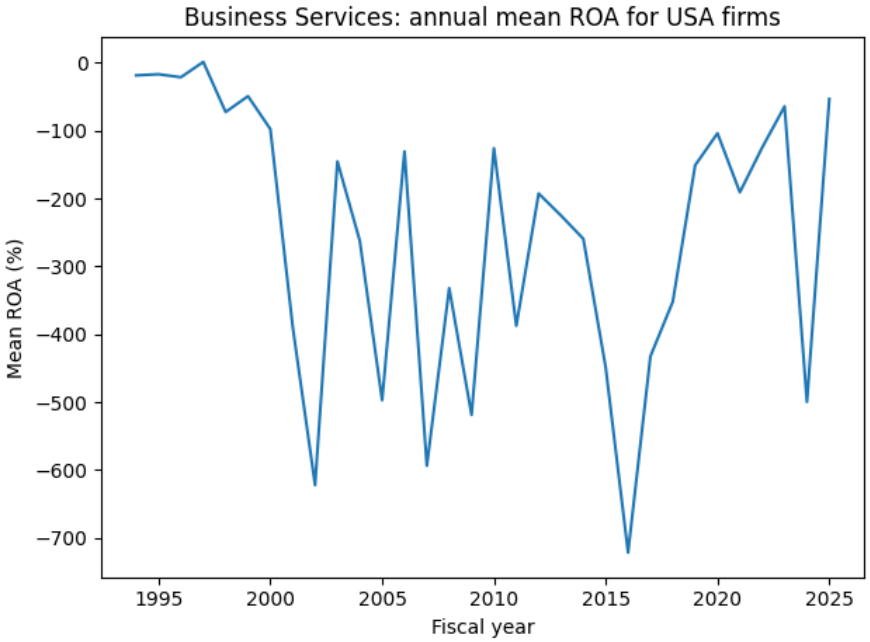


Figure 2 Annual mean ROA for USA firms

Steps C and D Text preparation and keyword methods

We apply the same cleaning to both years: lowercase text, replace punctuation with spaces and remove NLTK English stopwords. Original disclosures are retained. Alphabetic tokens are then used for keyword analysis and embeddings; standalone numbers are excluded. We do not stem or lemmatize, so customer and customers remain different terms.

```
text = str(text).lower()
text = re.sub(r"[^\w\s]|_", " ", text)
kept_words = []
for word in text.split():
    if word not in stop_words:
        kept_words.append(word)
cleaned_text = " ".join(kept_words)
```

This excerpt shows the cleaning logic inside the notebook's function. Replacing punctuation can split hyphenated expressions, and generic business terms remain even after stopword removal. These choices favor transparent, consistent preprocessing over complex language modeling.

Per firm keywords and sector coverage

The validated inner joins produce one disclosure per selected firm per year. There are 477 usable documents in 2020 and 510 in 2025, compared with 880 and 765 financial firms in those years. Approximate filing coverage is therefore 54.2% and 66.7%. Financial benchmarking retains firms without matched disclosures; the text sample is narrower.

Word counts rank the ten most frequent words within each disclosure. TF-IDF combines word frequency with inverse document frequency, reducing the influence of terms that appear widely across firms. Each disclosure is one document, with smoothed IDF and L2 normalization. Models are fitted separately by year. All firm-level rankings are computed, while the report shows an example to explain the distinction.

```
vectorizer = TfidfVectorizer(analyzer=word_tokens)
matrix = vectorizer.fit_transform(sample["item_1_clean"])
terms = vectorizer.get_feature_names_out()
```

Rank	Count term	Count	TF-IDF term	TF-IDF
1	worlds	33	worlds	0.646
2	inc	20	worldsplayer	0.271
3	technology	17	avatars	0.190
4	user	16	worldsshaper	0.190
5	company	14	bungie	0.162
6	patent	14	partes	0.145
7	patents	14	petitions	0.128
8	virtual	14	ptab	0.121
9	online	10	virtual	0.119
10	users	10	linden	0.108

Example: the supplied 2020 row labeled WORLDS INC. Worlds leads both rankings, while more distinctive names such as worldsplayer and avatars appear prominently under TF-IDF. Count and TF-IDF rankings answer different questions and should not be treated as interchangeable.

Step D Sector keywords and wordclouds

For sector frequency, we pool token counts across firms. For the TF-IDF cloud, we average each term's normalized score across all documents, including zeros. Pooled counts give longer documents more influence; mean TF-IDF gives a different view of term prominence. The two clouds use different weights, so their font sizes are not numerically comparable.

Rank	2020 term	Count	2025 term	Count
1	services	13395	data	14604
2	business	13322	services	13043
3	customers	10884	customers	11902
4	data	10308	business	11892
5	solutions	8966	company	10360
6	company	8793	platform	10032
7	products	7437	solutions	9342
8	may	7148	including	8763
9	including	7043	technology	7973
10	management	7036	also	7790

Rank	2020 mean TF-IDF term	2025 mean TF-IDF term
1	services	data
2	customers	services
3	business	customers
4	data	business
5	solutions	solutions
6	company	platform
7	products	company
8	platform	including
9	customer	customer
10	software	information

Seven of the ten most frequent terms recur across years, indicating continuity in core vocabulary. Services leads both 2020 rankings; data leads both in 2025. Platform enters the frequency top ten and rises from eighth to sixth in mean TF-IDF. These shifts describe disclosure emphasis; changes in document length and firm membership require the common-firm comparison below.



Figure 3 Word count and mean TF-IDF wordclouds for both years

Steps E and F Embeddings and year comparison

We train Word2Vec separately on retained disclosures across all industries, rather than only Business Services: 4,793 documents in 2020 and 4,509 in 2025, with vocabularies of 54,260 and 54,771 words. We use 100-dimensional vectors, a five-token window, minimum frequency five, five epochs, one worker and seed 508. Consecutive 1,000-token chunks accommodate long filings, losing some contexts across chunk boundaries.

```
model = Word2Vec(
    sentences=corpus, vector_size=100, window=5,
    min_count=5, workers=1, seed=508, epochs=5
)
```

We select services, software and customers from the wordclouds. Software is visible in both years and ranks tenth in 2020 mean TF-IDF, although it is less dominant than the other two terms. Word2vec places words used in similar local contexts near each other. The following lists show five nearest neighbors in descending cosine similarity; this score measures vector proximity, not a business probability.

Query	2020 neighbors	2025 neighbors
services	solutions, service, assistance, clients, ancillary	solutions, service, consulting, ancillary, networks
software	hardware, middleware, integrations, sdk, firmware	hardware, eda, middleware, saas, firmware
customers	clients, customer, users, consumers, buyers	clients, customer, users, consumers, buyers

Customers has the same five neighbors in both years, suggesting stable contextual associations in the supplied corpus. Services and software each retain three of five neighbors. The strongest software neighbor is hardware in both years, with cosine similarity of approximately 0.789. Neighbor changes concern cross-industry language and do not independently demonstrate sector adoption or competitive change.

Composition changes qualify the apparent transition

We normalize word counts by total alphabetic tokens and repeat the comparison among the 304 registrants appearing in both years. This holds membership constant, but still allows changes in disclosure length and business mix. Changes below are measured in percentage points (pp).

Term	Full sample change in pp	Common firms change in pp
data	+0.107	+0.031
platform	+0.090	−0.011
services	−0.149	−0.073
business	−0.201	−0.130

The data increase persists in the common-firm sample but becomes smaller. Platform reverses direction, indicating that sample composition matters to its apparent rise. A possible interpretation is increased emphasis on data-oriented offerings among continuing firms; disclosure practices and source-label issues remain alternative explanations. Separate TF-IDF fits have different IDF weights, and independently trained embedding vectors are not directly aligned, so we compare ranks and neighbor identities rather than raw vector differences.

Step G Microsoft financial peer benchmarking

We focus on one direction: financial benchmarking against software peers. Microsoft is compared with Oracle and Adobe, identified by GVKEY 012141, 012142 and 012540. All three have complete 2020–2025 records within SIC 73. Their product mix, asset requirements and fiscal calendars differ, so they provide benchmarks rather than an exhaustive or perfectly comparable competitor set.

Company	2025 sales	Revenue CAGR	2025 ROA	Asset turnover
MICROSOFT	281,724	14.52%	16.45%	0.455
ORACLE	67,357	10.72%	6.53%	0.257
ADOBE	23,769	13.06%	24.17%	0.806

The compound annual growth rate (CAGR) covers five intervals from 2020 to 2025. Microsoft grows fastest at 14.52%, compared with Oracle’s 10.72% and Adobe’s 13.06%. Its 2025 ROA is above Oracle’s but below Adobe’s. Adobe’s higher asset turnover is a benchmark, not a target that can be applied without accounting for business-model differences.

Microsoft represents 75.56% of the combined sales of these three firms, versus Oracle’s 18.07% and Adobe’s 6.37%. This is a selected-peer revenue share, not market share. A genuine market-share estimate would require a defined product market and its total revenue denominator.

The recent change is mainly in asset turnover

roa = net_income / assets
asset_turnover = revenue / assets
net_margin = net_income / revenue
Identity: ROA = net margin * asset turnover

Microsoft measure	2020	2022	2025
ROA	14.70%	19.94%	16.45%
Asset turnover	0.475	0.543	0.455
Net margin	30.96%	36.69%	36.15%

From the 2022 ROA peak to 2025, reported assets grow 19.27% annually while revenue grows 12.42%. ROA decreases 3.49 percentage points and asset turnover falls from 0.543 to 0.455, while margin changes only modestly. The arithmetic identity therefore points to weaker revenue generation per unit of reported assets as the main component of the recent ROA decline. It does not identify the underlying investment or establish causation.

The 2022–2025 window diagnoses a decline from a local peak. Microsoft's 2025 ROA remains above its 2020 value; the full period should not be described as a continuous deterioration.

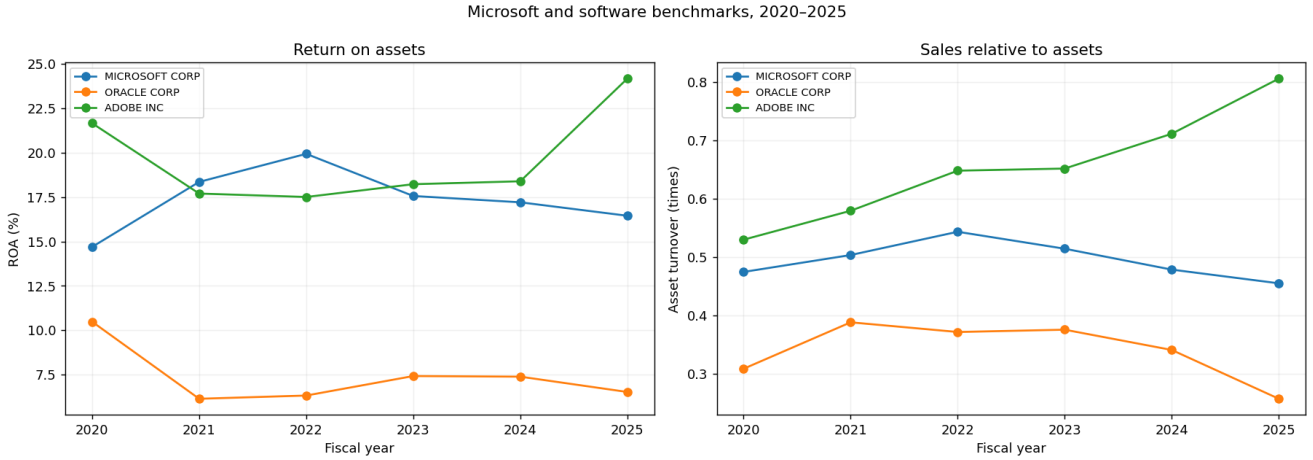


Figure 4 Microsoft and software benchmarks from 2020 to 2025

Step G One strategy suggestion

We recommend milestone-based capital allocation for major asset expansions. Before approving subsequent funding tranches, Microsoft should assess whether existing projects are meeting their planned utilization and revenue milestones. The gap between recent asset growth and revenue growth provides a reason to investigate deployment and demand assumptions, while the stable margin suggests the issue is not mainly a sharp margin contraction.

The proposed review would compare actual utilization and revenue with each project's business case. Where milestones are met, subsequent funding can proceed; where they are not, management can reassess the timing and scope of the next tranche. This is one funding-review recommendation, not a claim that aggregate ratios justify reducing all long-term investment.

Over the next two fiscal years, success would be assessed through project milestone attainment and stabilization of asset turnover, alongside commercially justified revenue growth and ROA. Targets should reflect project hurdle rates and investment horizons rather than copying Adobe's ratios. The supplied data does not contain utilization, segment performance, capital expenditure or asset composition, and it does not establish whether Microsoft already follows this process. These data would be needed before implementation.

Limitations and validation

Financial means are sensitive to outliers and changing coverage. Missing values are preserved, nonpositive assets are excluded from ROA and the very negative industry averages are reported transparently. Monetary units, currency comparability and fiscal-year conventions require confirmation before external use.

Filing availability differs between years, and CIK cardinality checks do not establish that supplied text belongs to the named firm. The longest-disclosure rule cannot identify the latest filing without accession or date metadata. The unexpected 2020 keyword cluster and identifier disagreements mean the NLP findings should remain exploratory until source identity is verified.

Keyword counts favor longer documents. Generic terms and separate singular/plural forms remain. Independent TF-IDF and word2vec fits can vary with vocabulary, sample coverage and software versions. The common-firm comparison reduces one source of bias but does not eliminate these limitations.

The final analytical notebook was run from a fresh namespace using complete supplied files. All 24 code cells completed without errors. Checks enforce unique financial firm-years, one-to-one filing joins, unique per-firm keyword ranks and the ROA identity. Successful execution confirms computational consistency, not the authenticity of the source data or a causal explanation.

Source provenance and AI assistance

Sources: BAIT 508 Group Project Public Firm Analysis v4.0; public_firms.csv; major_groups.csv; 2020_10K_item1_full.csv; and 2025_10K_item1_full.csv. Tables, numerical results and figures are calculated from these supplied datasets. No external company facts or currency conversions are used.

OpenAI's ChatGPT assisted with Python data loading and cleaning, SIC filtering, CIK joins, duplicate handling, word counts, TF-IDF, wordclouds and Word2Vec. It also supported troubleshooting, method explanations, integration of team contributions and report drafting. AI-assisted execution checks regenerated the analytical outputs. The authors remain responsible for reviewing source identity, code, calculations and final interpretations; successful execution does not resolve the documented source-data concerns.

AI tool reference: OpenAI. (2026). ChatGPT [Large language model]. https://chatgpt.com/